THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 13: **Our Environment** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** small fish	Zooplankton are primary consumers.
2	**(c)** eagle	Top consumer.
3	**(c)** separating biodegradable from non-biodegradable	For proper treatment.
4	**(c)** Plant, caterpillar, bird, hawk	Order of feeding.
5	**(a)** A hawk that eats snakes	Higher level.
6	**(c)** manure and clean water	Waste management.
7	**(c)** 50 J	Three transfers of 10%.
8	**(d)** Destroying the ozone layer	Chlorine destroys ozone.
9	**(c)** converting vegetable waste into manure	Natural decomposition.
10	**(b)** 10,000 J	10 x 1000.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Grass, grasshopper, frog, snake (1); Tertiary consumer: snake (1).	1+1
12	Alternate food sources exist (1); Loss of one species does not break the system (1).	1+1
13	DDT is non-degradable (1); Biomagnification along the food chain (1).	1+1
14	Primary consumer: 2000 J (1); Secondary consumer: 200 J (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	10 per cent law (1); Deer: 1200 J (1); Tiger: 120 J (1).	1+1+1
16	Pesticide gets into crops and water (1); Eaten by herbivores and accumulates (1); Carnivores accumulate more through biomagnification (1).	1+1+1
17	Decomposers act on dead matter at each level (1); Nutrients return to the producers (1); Materials are recycled (1).	1+1+1
	OR: Meat is at a higher trophic level (1); Energy is lost at each level (1); Plants are at the first level with more energy (1).	1+1+1
18	Non-biodegradable (1); They block drains, harm animals (1); Burning them releases toxic gases (1).	1+1+1
	OR: Energy falls by 90% at each step (1); Very little energy is left at the fifth level (1); Not enough to support another level (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Phytoplankton, zooplankton, small fish, large fish, fish-eating bird (1); T1: 100,000 J (1); T2: 10,000 J; T3: 1000 J (1); T4: 100 J; T5: 10 J (1); The top level (T5) has the least energy (1).	1+1+1+1+1
	OR: Sun gives energy to producers (1); Producers convert it into chemical energy (1); Passed on to consumers (1); Energy lost as heat at each step (1); Heat energy cannot be used by organisms, so it is not recycled (1).	1+1+1+1+1
20	Energy is lost at each level (1); Used in respiration, movement and body activity (1); Lost as heat (1); Only 10% stored in the body (1); So fewer organisms are supported at higher levels (1).	1+1+1+1+1
	OR: Producers: base of the food chain (1); Herbivores: link producers to carnivores (1); Carnivores: control herbivore numbers (1); Decomposers: recycle nutrients (1); Removal of one disturbs the balance of all (1).	1+1+1+1+1

Mistakes to avoid:

- Drawing arrows in a food chain from the eater to the food; the arrow points from the organism that is eaten to the one that eats it.
- Writing that 90% of the energy passes on to the next level; only 10% is transferred and 90% is lost, so energy flows in one direction and does not cycle.

- Counting decomposers as a trophic level of the food chain, or calling the Sun a producer; green plants are producers, the Sun is the source of energy.
- Confusing ozone (O_3) with oxygen (O_2), and writing that CO_2 (instead of CFCs) is the main cause of ozone depletion.
- Saying that all waste is non-biodegradable or that plastics decompose quickly; plastics, glass and DDT are non-biodegradable.
- Explaining biomagnification as increase of pesticide in producers only; the concentration is highest in the **top** consumers.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 13: **Our Environment** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** UV rays of the sun	Protects from skin cancer.
2	**(c)** Vegetable peels	Biodegradable waste.
3	**(b)** Grass	Producers form the first trophic level.
4	**(a)** Antarctica	In 1985.
5	**(a)** save trees	Conserves resources.
6	**(d)** recycle nutrients	They return materials to the soil.
7	**(c)** O2	The diatomic form.
8	**(d)** Using cloth bags	Reduces plastic.
9	**(d)** All organisms	Interdependence.
10	**(a)** UNEP	In 1987.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Ultraviolet radiation (1); Skin cancer (1).	1+1
12	Lost in respiration and as heat (1); Only 10% is passed on (1).	1+1
13	Energy available is very low at the top level (1); Only a small population can be supported (1).	1+1
14	They are stable gases that rise (1); They are broken by UV to release chlorine (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Plant foods are at the first level (1); Less energy lost from fewer transfers (1); More people can be fed from the same land (1).	1+1+1
16	Biodegradable waste is decomposed by microbes (1); Forms manure (1); Reduces waste and improves the soil (1).	1+1+1
17	CFCs release chlorine in the stratosphere (1); Chlorine destroys ozone (1); Ozone depletion harms life (1).	1+1+1
	OR: No food is made (1); Consumers starve (1); The whole ecosystem collapses (1).	1+1+1
18	Waste is converted into new products (1); Saves raw materials (1); Reduces the amount of garbage (1).	1+1+1
	OR: Biodegradable waste can be composted (1); Non-biodegradable waste can be recycled (1); It reduces pollution (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	DDT enters the lake from run-off (1); Taken by the plankton (1); Concentrated in fish (1); More concentrated in the fish-eating birds (1); Humans at the top get the highest dose (1).	1+1+1+1+1
	OR: Advantage: protects crops (1); Disadvantage: harms useful organisms (1); Non-degradable pesticides undergo biomagnification (1); Pollutes water and soil (1); Alternatives: organic farming, biological control, crop rotation (1).	1+1+1+1+1
20	Ozone is O3, found in the stratosphere (1); UV splits O2 into atoms which combine with O2 to form O3 (1); CFCs release chlorine that breaks ozone (1); More UV reaches the earth causing skin cancer (1); UNEP framework to freeze CFC production in 1987 (1).	1+1+1+1+1
	OR: Cloth bags reduce plastic (1); Reusable bottles reduce waste (1); Compost converts kitchen waste into manure (1); Recycled paper saves trees (1); CFC-free refrigerators protect the ozone layer (1).	1+1+1+1+1

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 13: **Our Environment** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** infertile	It stays unchanged.
2	**(a)** increase the frog population	Predators control populations.
3	**(a)** pollution and biomagnification	Harmful.
4	**(b)** toxic gases	Pollutes air.
5	**(a)** more than one trophic level	E.g. an omnivore.
6	**(d)** non-biodegradable	Accumulate.
7	**(d)** Aluminium foil	It does not decay easily.
8	**(d)** second	Herbivore position.
9	**(c)** biomagnification	Accumulation.
10	**(a)** reduce its production	Reduce at source.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Grazing chain starts with green plants (1); The decomposer chain starts with dead matter (1).	1+1
12	They decompose dead organisms and wastes (1); They return minerals to the soil (1).	1+1
13	They are biodegradable (1); Reduce pollution (1).	1+1
14	They pollute soil and water (1); They enter the food chain and get biomagnified (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Grass grows in excess (1); The tiger loses its prey and its numbers fall (1); The balance of the ecosystem is disturbed (1).	1+1+1
16	Provides alternate food sources (1); Organisms can shift diets (1); The system is balanced (1).	1+1+1
17	Phytoplankton: producer (1); Zooplankton: primary consumer; fish: secondary consumer (1); Bird: tertiary consumer (1).	1+1+1
	OR: Cause: CFCs (1); More UV rays reach the earth (1); Effects: skin cancer, cataract and crop damage (1).	1+1+1
18	Producers: autotrophic (1); Consumers: heterotrophic (1); Producers use solar energy; consumers use food (1).	1+1+1
	OR: Using CFCs (1); Excess use of plastics (1); Use of pesticides and improper waste disposal (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Grass eaten by grasshopper and rabbit (1); Grasshopper eaten by frog and bird (1); Rabbit eaten by snake and hawk (1); Frog eaten by snake; snake by hawk (1); If one species is lost, others sustain the web (1).	1+1+1+1+1
	OR: Natural: self-sustaining, e.g. forest, pond (1); Artificial: needs human care, e.g. crop field, aquarium (1); Natural has high biodiversity (1); Artificial has low biodiversity (1); Energy source is the Sun in both (1).	1+1+1+1+1
20	Non-biodegradable materials remain for years (1); Plastics clog drains and harm animals (1); DDT biomagnifies (1); Plan: separate dustbins, compost pit, no plastic (1); Reuse paper; recycle bottles (1).	1+1+1+1+1
	OR: In a grassland chain, removal of snakes increases frogs (1); More frogs reduce insects (1); Insects increase and damage grass (1); The balance is disturbed (1); So every organism has a role (1).	1+1+1+1+1

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